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Grid-Scale Battery Storage Monthly Report, October 2026

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The price of electricity changes with the time of day it is used. A "grid-scale battery" (a large battery connected to the power grid that stores electricity and releases it) earns money from that price difference. This monthly report uses published figures and our own records to show this month's trends in charts.

1. How much did one 2 MW / 8 MWh battery site earn in October 2026? (by grid area, estimate)

The highest income estimates for October were Chubu at ¥10.43 million, Hokkaido at ¥8.07 million and Kyushu at ¥7.45 million. The lowest was Tokyo at ¥5.30 million.

Grid area	Electricity trading (price spread, October)	Balancing market (Tertiary reserve 2, September)	Capacity market (per month)	Income estimate (¥10,000/month)	What is added
Hokkaido	557	341	250	807	Capacity + spread
Tohoku	458	170	252	710	Capacity + spread
Tokyo	278	131	252	530	Capacity + spread
Chubu	347	837	206	1,043	Capacity + balancing
Hokuriku	349	147	206	555	Capacity + spread
Kansai	340	323	206	546	Capacity + spread
Chugoku	358	90	206	564	Capacity + spread
Shikoku	336	68	206	542	Capacity + spread
Kyushu	368	493	252	745	Capacity + balancing

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All figures are in ¥10,000 per month. Because of rounding, adding up the table figures can differ from the income estimate by ¥10,000. The table in section 6 shows which figures were multiplied and how.

We estimated, for each of the 9 grid areas, how much income one battery site with 2 MW output and 8 MWh capacity would have earned in October 2026. The income estimate is the capacity market amount plus the larger of electricity trading (price spread) and the balancing market. The price spread and the balancing market compete for the same 2 MW, so we do not add both. It is an income estimate before costs such as aggregator fees, maintenance and taxes. It does not show or guarantee actual earnings. The balancing market uses September figures (September 1–29). It assumes the site won in every 30-minute slot that had winning bids, so it comes out higher than reality (section 6). The capacity market amount is the clearing price of the latest auction that a battery site built today would enter (FY2029), divided by 12 months.

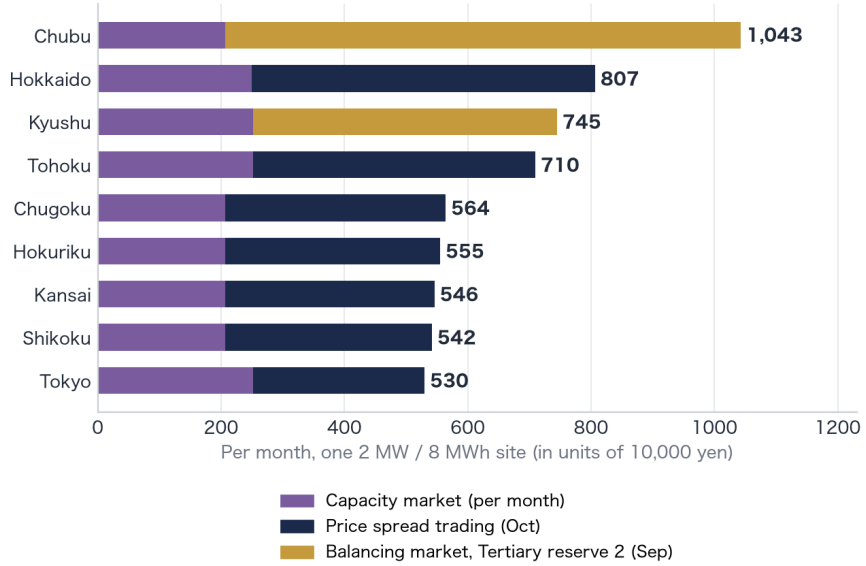


Figure 1 Income estimate for one 2 MW / 8 MWh battery site in October 2026 (by grid area, estimate, in units of ¥10,000 per month)

Source: Our estimate based on the JEPX spot market, EPRX preliminary figures and clearing prices published by OCCTO. Calculations in section 6, assumptions in section 12.

2. Three trends this month

Wholesale electricity is electricity bought and sold by power companies and retailers. Its price is set on the "spot market" of the Japan Electric Power Exchange (JEPX) (a market where electricity for the next day is traded the day before).

- The average wholesale electricity price in October was 27.02 yen/kWh in Tokyo (same month last year 13.06 yen/kWh, a difference of 13.96 yen) and 15.07 yen/kWh in Kyushu (same month last year 10.74 yen/kWh, a difference of 4.34 yen).
- Within the day, the cheapest average was around 12:00 in both Tokyo and Kyushu (Tokyo 19.03 yen, Kyushu 4.94 yen). The highest was 37.11 yen around 16:30 in Tokyo and 24.74 yen around 17:30 in Kyushu.
- In our estimate for one battery site in Tokyo, the monthly figures are ¥2.78 million from the electricity price spread (October), ¥1.31 million from Tertiary reserve 2 in the balancing market (September) and ¥2.52 million from the capacity market (FY2029 price).

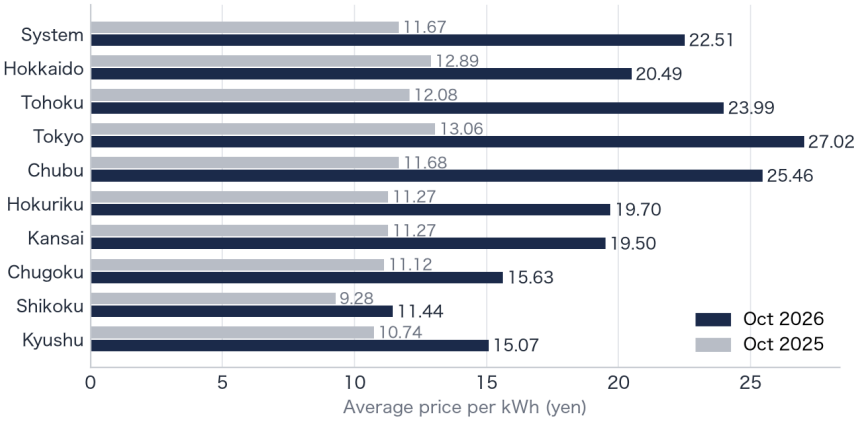


Figure 2 Average wholesale electricity price (yen/kWh). Navy is October 2026, grey is the same month a year earlier

Source: Spot market trading results published by the Japan Electric Power Exchange (JEPX). Simple average of prices for each 30 minutes. Our tally.

3. How the price of electricity moves within a day

The price of electricity changes every 30 minutes. The chart shows the shape of a day, averaging prices at the same time across the month. If a battery stores electricity when it is cheap and releases it when it is expensive, the difference becomes the battery's income.

The time-of-day averages ranged from 19.03 yen around 12:00 to 37.11 yen around 16:30 in Tokyo, and from 4.94 yen around 12:00 to 24.74 yen around 17:30 in Kyushu.

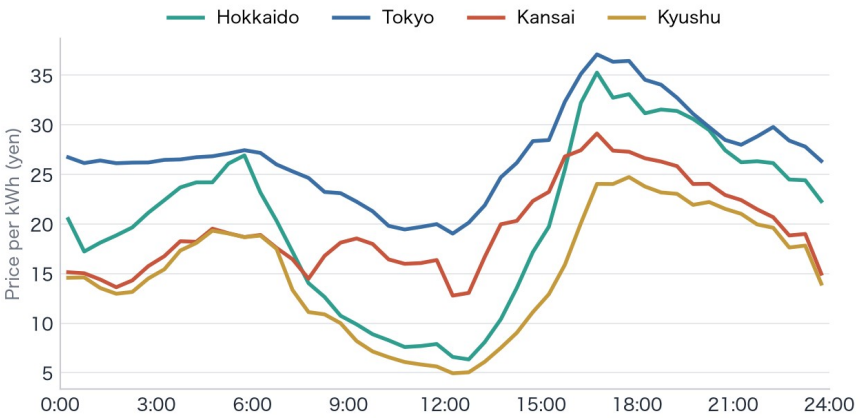


Figure 3 Peaks and valleys of the daily price (October 2026 average, yen/kWh)

Source: JEPX spot market. Our tally.

4. There is a high-low gap every day

We lined up, day by day, the difference between the highest and lowest price within each day. The larger the gap on a day, the more room a battery has to earn from the price spread.

The monthly average of the daily high-low gap was 36.50 yen in Hokkaido, 24.05 yen in Kansai, 23.15 yen in Kyushu and 21.87 yen in Tokyo. Slots priced at 0.01 yen occurred 91 times in Kyushu and 16 times in Tokyo.

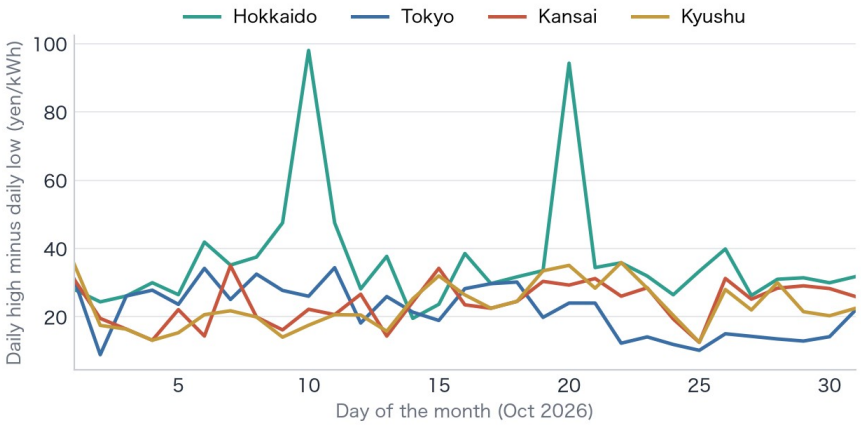


Figure 4 Daily price gap (the day's highest price − lowest price, yen/kWh)
Source: JEPX spot market. Our tally.

5. Three income sources for a battery site

A battery site has three main sources of income. The electricity price spread (wholesale power market), the balancing market (a market for buying and selling the "ability" to fix surpluses or shortages of electricity), and the capacity market (a market that buys future electricity supply capacity in advance). In the capacity market, power plants and battery sites receive a fixed income per kW.

1 Price spread	2 Balancing market	3 Capacity market
Wholesale market (JEPX)	Balancing exchange (EPRX)	Grid coordinator (OCCTO)
Buy and store power when cheap, sell it when it is expensive.	Sell ahead the ability to fix surpluses and shortages.	The state buys future supply capacity in advance.
Price set a day ahead **Income changes day to day**	**Price per 30 minutes** **Income by kW offered**	**Yearly auction, ~4 years ahead** **Fixed income per kW**

Figure 5 Three income sources for a battery site
Source: Rules of the Japan Electric Power Exchange (JEPX), the Electric Power Reserve Exchange (EPRX) and the Organization for Cross-regional Coordination of Transmission Operators (OCCTO).

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6. How much does a battery earn? (estimate)

These are monthly figures for one battery site with 2 MW output and 8 MWh capacity. For electricity trading, we assume it charges in the cheapest 4 hours of the day and discharges in the most expensive 4 hours. For the balancing market, we assume it keeps offering 2 MW in Tertiary reserve 2. For the capacity market, the amount is the clearing price × 2,000 kW ÷ 12. All are rough estimates. They do not show or guarantee actual earnings (assumptions in section 12). The gold bars for the balancing market are September figures (September 1–29), a different month from the navy bars for October.

In our estimate for one battery site, Tokyo had ¥2.78 million from the price spread (October), ¥1.31 million from Tertiary reserve 2 (September) and ¥2.52 million from the capacity market (per month). Kyushu had ¥3.68 million, ¥4.93 million and ¥2.52 million respectively.

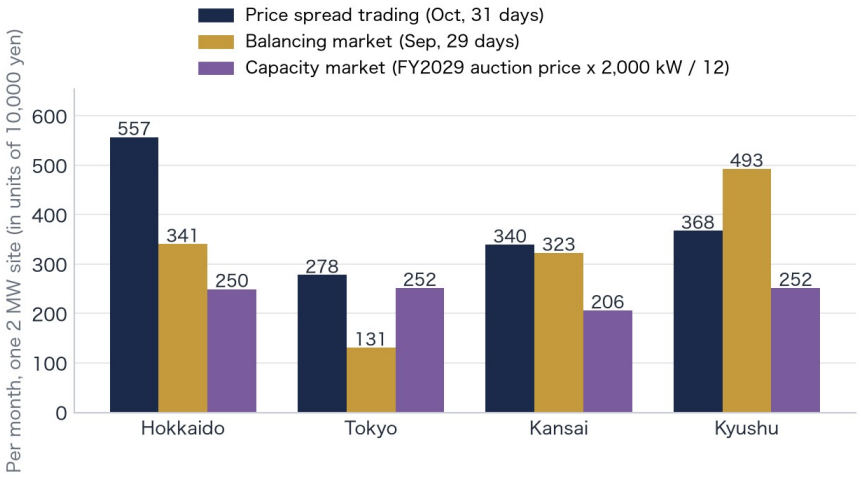


Figure 6 Earnings estimate for one 2 MW / 8 MWh battery site (estimate, in units of ¥10,000 per month)

Source: Our estimate based on the JEPX spot market, EPRX preliminary figures and clearing prices published by OCCTO. Assumptions in section 12.

The figures in Figures 1 and 6 come from the following calculations.

- Amount bought and stored per day: capacity 8,000 kWh × depth of discharge 95% = 7,600 kWh. It buys in the 8 cheapest 30-minute slots of the day (4 hours).
- Amount sold per day: 7,600 kWh × round-trip efficiency 85% = 6,460 kWh. It sells in the 8 most expensive 30-minute slots of the day (4 hours).
- ① Average price of the 8 cheap slots (yen/kWh, average of October's daily values), ② average price of the 8 expensive slots (yen/kWh).
- ③ Daily price spread balance = 6,460 kWh × ② − 7,600 kWh × ① (yen). ④ Electricity trading = ③ × 31 days (¥10,000 per month).
- ⑤ Average winning price for Tertiary reserve 2 (yen/kW per 30 min, September), ⑥ number of slots with winning bids (out of 29 days × 48 slots = 1,392 slots). ⑦ Balancing market = 2,000 kW × ⑤ × ⑥ (¥10,000 per month).
- ⑧ Capacity market clearing price (yen/kW per year, FY2029). ⑨ Capacity market = 2,000 kW × ⑧ ÷ 12 (¥10,000 per month).
- ⑩ Income estimate = ⑨ + (the larger of ④ and ⑦) (¥10,000 per month).

Grid area	①	②	③	④	⑤	⑥	⑦	⑧	⑨	⑩
Hokkaido	6.80	35.82	179,732	557.17	1.963	868	340.78	14,972	249.53	806.70
Tohoku	10.64	35.38	147,683	457.82	4.389	194	170.31	15,111	251.85	709.67
Tokyo	18.41	35.55	89,726	278.15	1.004	653	131.09	15,111	251.85	530.00
Chubu	14.74	34.65	111,835	346.69	8.317	503	836.70	12,388	206.47	1,043.16
Hokuriku	9.98	29.16	112,518	348.81	6.263	117	146.56	12,388	206.47	555.27
Kansai	9.95	28.67	109,537	339.56	5.017	322	323.11	12,388	206.47	546.03
Chugoku	6.03	24.96	115,389	357.71	1.179	382	90.06	12,388	206.47	564.17
Shikoku	2.92	20.19	108,227	335.50	0.384	885	67.91	12,388	206.47	541.97
Kyushu	5.38	24.68	118,552	367.51	4.450	554	493.03	15,112	251.87	744.89

The intermediate figures in the table are rounded, so multiplying them on a calculator may not match to the yen. Electricity trading and the balancing market use the same 2 MW, so the site cannot earn the full amount from both at once. That is why the income estimate adds only the larger of the two to the capacity market.

The balancing market estimate assumes the site won at the average winning price in every slot that had winning bids. In September, 26.9% of bids for Tertiary reserve 2 were accepted nationwide (table in section 11). The amount a site can actually win will be smaller than this.

7. Batteries are also used in the balancing market

In the balancing market, the lower the product number, the faster the response required. The chart shows the share of batteries in the volume awarded. Published figures for October 2026 are not yet in our tally, so we show September 2026 (preliminary), the latest month with all products available.

In September 2026 (preliminary), the battery share of awarded volume was 27.9% for Primary reserve, 10.4% for Composite product, 1.8% for Secondary reserve 1 and 0.8% for Tertiary reserve 2.

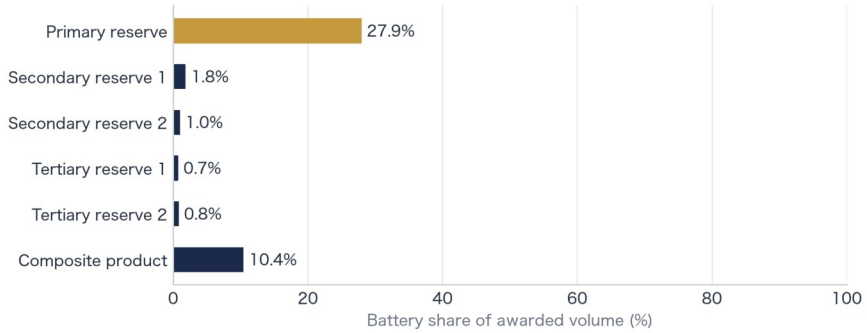


Figure 7 Share of batteries in the balancing market (%)

Source: Trading results by power source type (preliminary) published by the Electric Power Reserve Exchange (EPRX). Our tally.

8. The capacity market buys future supply capacity

The capacity market buys electricity supply capacity four years ahead through an auction once a year. The price is set in yen per kW per year. The left chart shows past main auction clearing prices. The right chart shows the battery and pumped hydro capacity contracted in the Long-Term Decarbonization Power Source Auction (a separate auction that supports new builds such as batteries and pumped hydro). Results are published once a year, so they do not change every month. The latest is FY2029 (published January 2026).

These are the latest figures, published once a year. In the capacity market, Tokyo was 3,495 yen/kW for FY2025 and 15,111 yen/kW for FY2029. In the Long-Term Decarbonization Power Source Auction, batteries and pumped hydro in the FY2025 auction were 819 MW lithium-ion and 886 MW non-lithium.

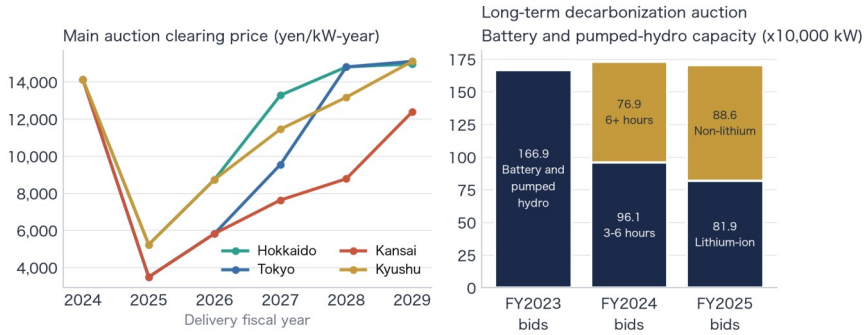


Figure 8 Capacity market clearing prices (left, yen/kW) and battery and pumped hydro capacity contracted in the Long-Term Decarbonization Power Source Auction (right, in units of 10,000 kW)

Source: Auction results and capacity contract documents published by the Organization for Cross-regional Coordination of Transmission Operators (OCCTO) (sources in section 12). As published.

9. Where are battery projects growing?

We counted the projects newly listed this month on our DealRoom (our data room for buyers, where we list battery site projects), by grid operator's grid area. These are figures for projects we handled, not counts for the whole market. Transfer of rights means selling the rights to the land and power. Completed-build handover means selling once the facility is built.

New listings in October were 77 projects, 309.1 MW. By grid area: Tokyo 33 projects, 99.8 MW; Tohoku 8 projects, 73.4 MW; Chubu 11 projects, 69.9 MW.

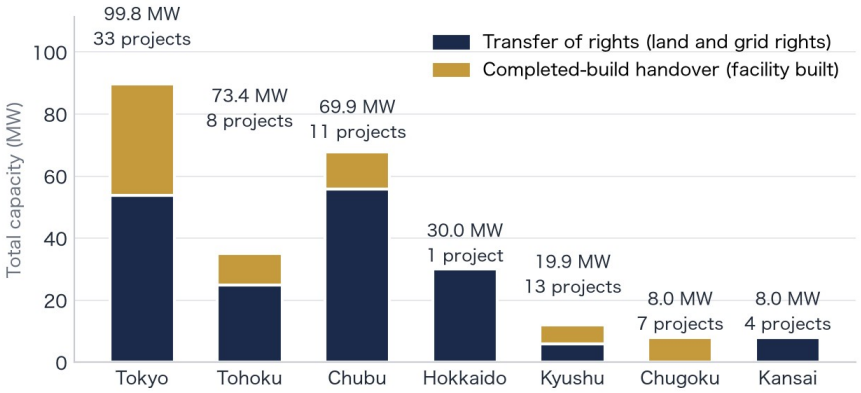


Figure 9 Output (MW) of projects listed on our DealRoom in October 2026

Source: Our DealRoom records (project locations shown only to grid area level; no prices or parties).

10. How many months and how much does it take to connect to the grid?

A grid connection study is a procedure where the grid operator checks "can this site connect, and how much will the work cost?" The grid connection cost contribution is the part of the connection work cost paid by the applicant. From the responses we received, we tallied only high-voltage cases (connecting to 6,600 V distribution lines) by grid area. Extra-high voltage is not included. Grid areas with fewer than 5 cases are not shown.

As of October 19, 2026, the high-voltage medians were 8.0 months and ¥3.02 million in Tokyo, 12.5 months and ¥12.50 million in Kyushu, and 17.0 months and ¥32.35 million in Tohoku.

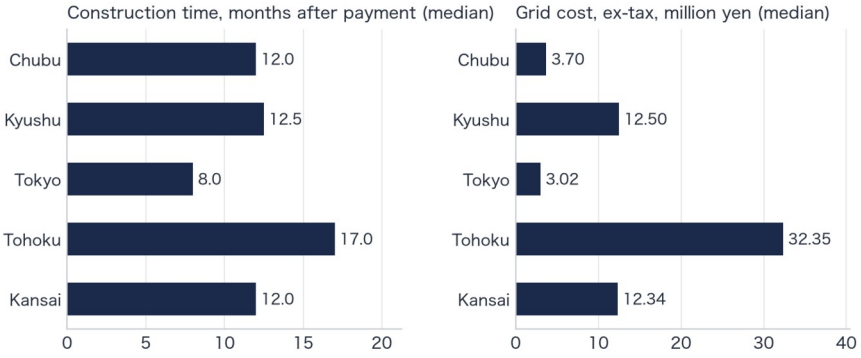


Figure 10 Median grid connection study responses (high voltage) by grid area (left: months of construction, right: grid connection cost contribution, excluding tax)

Source: Our tally of high-voltage grid connection study responses received from grid operators (as of October 19, 2026).

11. Market figures (tables ready to use in your materials)

We put the figures behind the charts into tables you can use in your materials. They cover the wholesale electricity market (JEPX), figures for each balancing market product (bid ratio, award rate, price, battery share) and capacity market clearing prices.

JEPX spot market (October 2026, by area, yen/kWh)

Area	Monthly average	Same month last year	Daytime (8:00–16:00)	Evening (16:00–20:00)	Daily high-low gap (monthly average)	Highest	Slots at 30 yen or more	Slots at 0.01 yen
System	22.51	11.67	16.02	31.06	25.16	47.88	220	39

Area	Monthly average	Same month last year	Daytime (8:00–16:00)	Evening (16:00–20:00)	Daily high-low gap (monthly average)	Highest	Slots at 30 yen or more	Slots at 0.01 yen
Hokkaido	20.49	12.89	11.31	32.25	36.50	105.00	303	67
Tohoku	23.99	12.08	15.78	34.19	28.92	50.01	437	95
Tokyo	27.02	13.06	23.12	34.69	21.87	50.01	466	16
Chubu	25.46	11.68	20.38	33.55	24.42	46.45	387	24
Hokuriku	19.70	11.27	18.29	26.92	24.23	46.45	218	24
Kansai	19.50	11.27	18.22	26.76	24.05	46.45	215	29
Chugoku	15.63	11.12	9.86	23.10	23.07	44.79	63	79
Shikoku	11.44	9.28	7.50	16.61	21.12	37.51	23	146
Kyushu	15.07	10.74	8.31	23.10	23.15	44.79	53	91

Trading volume (October 2026, total for 31 days): total contracted volume 30,319 million kWh (same month last year 22,279 million kWh), sell bids 45,218 million kWh, buy bids 38,191 million kWh.

Source: Spot market trading results published by the Japan Electric Power Exchange (JEPX). Prices are simple averages of 30-minute slots. Our tally.

Balancing market (September 2026, preliminary, nationwide)

Product	Volume sought (MW)	Volume bid (MW)	Volume awarded (MW)	Bid ratio (bid ÷ sought)	Award rate (awarded ÷ bid)	Average winning price (yen/kW per 30 min)	Battery share
Primary reserve	1,388.7	1,710.0	1,337.3	123.1%	78.2%	3.43	27.9%
Primary reserve (offline)	497.8	418.4	265.7	84.0%	63.5%	5.79	—
Secondary reserve 1	962.6	1,996.2	1,102.3	207.4%	55.2%	2.49	1.8%
Secondary reserve 2	983.1	3,709.3	2,163.2	377.3%	58.3%	2.51	1.0%
Tertiary reserve 1	2,498.5	5,895.9	3,021.0	236.0%	51.2%	2.28	0.7%
Tertiary reserve 2	376.3	1,371.7	368.9	364.5%	26.9%	2.22	0.8%
Composite product	3,383.8	6,641.4	3,623.5	196.3%	54.6%	2.64	10.4%

Volumes are averages per 30-minute slot. Products with a bid ratio below 100% did not receive enough bids for the volume sought. The award rate is the share of the volume bid that was awarded. The average winning price is weighted by volume awarded.

Source: Trading results (preliminary) and trading results by power source type published by the Electric Power Reserve Exchange (EPRX). Our tally.

Capacity market main auction area prices (yen/kW per year)

Delivery fiscal year	Hokkaido	Tohoku	Tokyo	Chubu	Hokuriku	Kansai	Chugoku	Shikoku	Kyushu
FY2024	14,137	14,137	14,137	14,137	14,137	14,137	14,137	14,137	14,137
FY2025	5,242	3,495	3,495	3,495	3,495	3,495	3,495	3,495	5,242
FY2026	8,749	5,833	5,834	5,832	5,832	5,832	5,832	5,832	8,748
FY2027	13,287	9,044	9,555	7,823	7,638	7,638	7,638	7,638	11,457
FY2028	14,812	14,812	14,812	10,280	8,785	8,785	8,785	8,785	13,177
FY2029	14,972	15,111	15,111	12,388	12,388	12,388	12,388	12,388	15,112

Battery and pumped hydro capacity contracted in the Long-Term Decarbonization Power Source Auction (10,000 kW)

Auction fiscal year	Category	Contracted capacity (10,000 kW)
FY2023 auction	Batteries and pumped hydro	166.9
FY2024 auction	3–6 hours	96.1
FY2024 auction	6 hours or more	76.9
FY2025 auction	Lithium-ion	81.9
FY2025 auction	Non-lithium	88.6

Source: Auction results and capacity contract documents published by the Organization for Cross-regional Coordination of Transmission Operators (OCCTO) (sources in section 12). As published.

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12. Estimate assumptions and terms

- The estimates are rough figures we calculated from published prices and the assumptions below. They do not show or guarantee actual earnings.
- Battery site: 2 MW output, 8 MWh capacity, round-trip efficiency 85%, depth of discharge 95%. Maximum 2 MW for both charging and discharging.
- Electricity trading: of the day's 30-minute slots, it charges in the 8 cheapest slots and discharges in the 8 most expensive. The charge amount is capacity × depth of discharge, and the discharge amount is 85% of that. We do not account for slot order, state-of-charge limits, the effect of our own trades on the market, or fees, taxes and grid costs.
- Balancing market: the total if the site offered 2 MW in Tertiary reserve 2 in every slot and won at the average winning price (yen/kW per 30 min) of the area where the power source is located in that slot. Slots with no winning bids count as 0. In reality, there are slots where the site would lose to competition, and offering reserve leaves times when the battery cannot be used for trading, so this assumption comes out higher than reality.
- Capacity market: the latest main auction (FY2029) clearing price (area price, yen/kW per year) × rated output 2000 kW ÷ 12. The income is received in that delivery fiscal year. It does not account for the coefficient used to rate supply capacity, deductions under transitional measures, or reductions from effectiveness tests or assessments. Income from the Long-Term Decarbonization Power Source Auction is not included.
- Months covered: electricity trading uses October (31 days), the balancing market uses September (September 1 –29). The months differ until the October 2026 balancing market figures are included in our tally.
- Figures are shown as they appear in the sources. We do not add up multiple months or convert to annual figures (the monthly capacity market amount is the annual clearing price divided by 12).
- Average prices are simple averages of 30-minute slots (not weighted by trading volume). Daytime means slots from 8:00 to 16:00, evening means slots from 16:00 to 20:00.

Sources (OCCTO, capacity market results): FY2024 https://www.occto.or.jp/iinkai/youryou/kentoukai/2020/files/youryou_kentoukai_27_05.pdf / FY2025 https://www.occto.or.jp/iinkai/youryou/kentoukai/2021/files/youryou_kentoukai_35_03_1.pdf / FY2026 https://www.occto.or.jp/iinkai/youryou/kentoukai/2022/files/youryou_kentoukai_43_03.pdf / FY2027 https://www.occto.or.jp/market-board/market/oshirase/2023/files/240124_mainauction_youryouyakujokekka_kouhyou_jitsujukyu2027.pdf / FY2028 https://www.occto.or.jp/market-board/market/oshirase/2024/files/250129_mainauction_youryouyakujokekka_kouhyou_jitsujukyu2028.pdf / FY2029 https://www.occto.or.jp/assets/news/capacity-market/260120_mainauction_youryouyakujokekka_kouhyou_jitsujukyu2029.pdf

Sources (OCCTO, Long-Term Decarbonization Power Source Auction): FY2023 auction https://www.occto.or.jp/assets/260605_auction_keiyakujyoukyou_kouhyou.pdf / FY2024 auction https://www.occto.or.jp/market-board/market/jitsujukyukanren/files/longauction_keiyakukekka_2024.pdf / FY2025 auction https://www.occto.or.jp/assets/various/capacity-market/jitsujukyukanren/yoryokakuhokeiyaku/longauction_keiyakukekka_2025.pdf

- Capacity market results for each delivery fiscal year are published in a different year (the FY2024 results in 2020, the FY2029 results in January 2026). The number of battery projects in the Long-Term Decarbonization Power Source Auction is not shown, because it is not in the published summary tables.

Sources: Japan Electric Power Exchange (JEPX) spot market trading results / Electric Power Reserve Exchange (EPRX) trading results (preliminary) / Organization for Cross-regional Coordination of Transmission Operators (OCCTO) capacity market main auction results and Long-Term Decarbonization Power Source Auction capacity contract documents / our tally of grid connection study responses we received / our DealRoom records.

Terms: The grid = the power companies' transmission network. Grid connection = connecting to the transmission network. MW = a measure of power (1 MW = 1,000 kW). MWh = the amount of electricity that can be stored. yen/kWh = the price of 1 kWh of electricity.

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